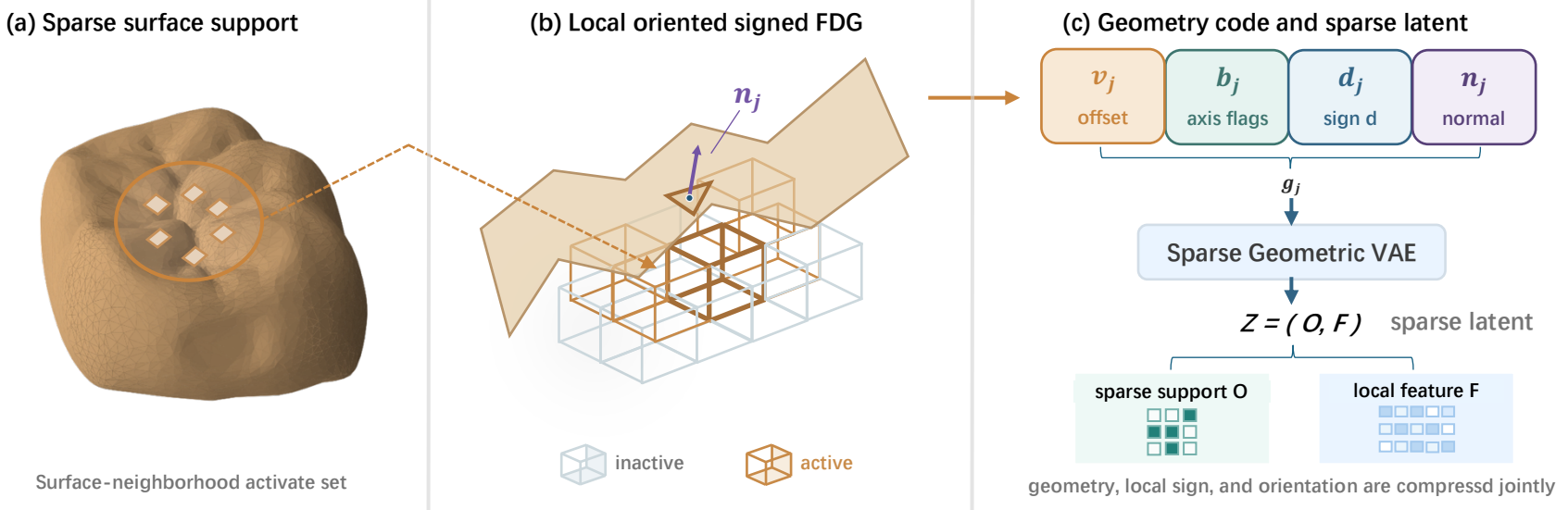

(a) Sparse surface support
Surface-neighborhood activate set
(b) Local oriented signed FDG
n_j
inactive
active
(c) Geometry code and sparse latent
v_j
offset
b_j
axis flags
d_j
sign d
n_j
normal
g_j
Sparse Geometric VAE
$Z = (O, F)$
sparse latent
sparse support O
local feature F
geometry, local sign, and orientation are compressd jointly